Battery Health Monitor

Component guide: what each part is, what it does in the rig, and why it was chosen

The system estimates a battery's **State of Health (SOH)** – how much charge it can still hold compared with new – and its **Remaining Useful Life (RUL)**. It does this by watching one complete discharge of a real battery pack and feeding four measurements from that discharge into machine-learning models trained on NASA's Li-ion battery ageing data.

Stage	What happens	Parts involved
1. Store energy	Three 18650 cells in series hold the charge being tested.	Cells, holders, BMS
2. Discharge	Flipping the switch sends about 1.5 A through two power resistors until the pack is empty.	Switch, resistors
3. Measure	The shunt turns current into a tiny voltage; the PZEM-017 measures voltage and current.	Shunt, PZEM-017
4. Transmit	The ESP32 reads the PZEM over RS485 every 5 s and sends the readings to the laptop over WiFi.	MAX485, divider, ESP32
5. Analyse	The backend detects the discharge, computes features and runs the models.	Laptop, backend, ML models
6. Display	The dashboard shows live readings, the predicted SOH/RUL and the measured SOH.	Streamlit dashboard

The two design rules that explain most choices
1. Match the training data. The models learned from single 2 Ah 18650 Li-ion cells discharged at about 2 A, starting full and running to empty. The closer the rig matches that, the more meaningful the prediction.
2. Keep the PZEM-017 powered. It needs at least 7 V on its measuring side. One cell (3.7 V) is too low, which is why the rig uses three cells in series (8.1–12.6 V).

A. The battery under test

1. 18650 Li-ion cells (×3)

≈ 2000 mAh, 3.7 V nominal, 4.2 V full, 2.7 V empty

What it is	Cylindrical rechargeable lithium-ion cells, 18 mm × 65 mm – the same type used in laptop batteries.
Its job here	They are the battery whose health the system estimates. Good (new) cells and worn cells are tested separately to show the system can tell them apart.
Why this one	The ML models were trained on NASA's 18650 Li-ion cells of about 2 Ah. Using the same type and capacity keeps the four input features within the range the models learned. A different chemistry (e.g. LiFePO4 or VRLA) or capacity would push the inputs outside that range and make the prediction meaningless.

2. 3S configuration (cells in series)

3 × 4.2 V = 12.6 V full, 3 × 2.7 V = 8.1 V empty

What it is	The three cells are connected + to − so their voltages add, while the same current flows through all of them.
Its job here	Raises the pack voltage so the PZEM-017 can power itself. The firmware then divides voltage, power and energy by 3 (*CELLS_IN_SERIES* = *3*), so the models see one 'average cell' – the same scale as the training data.
Why this one	A single cell (max 4.2 V) is below the PZEM-017's 7 V minimum, which caused the fake 4.9 V readings and communication errors seen earlier. Two cells drop below 7 V before empty. Three cells stay above 7 V for the whole discharge. In series, current and capacity (mAh) are the same as one cell, so no extra correction is needed for them.

3. Single 18650 battery holders

3 for the pack (+ spare holders for charging)

What it is	Plastic cradles with spring contacts and wire leads that hold one cell each.
Its job here	Hold the three cells and let them be joined in series with wires; the joins between holders give easy connection points for the BMS balance wires.
Why this one	Soldering directly onto cells can damage them and is a fire risk. Separate holders make it easy to remove cells for charging and easy to reach the joins between cells, which a single 3-slot holder hides.

4. 3S BMS protection board (20 A)

for 3-cell Li-ion, 12.6 V; pads 0V/B−, 4.2V, 8.4V, 12.6V/B+, P−

What it is	A Battery Management System board: a small circuit with MOSFET switches that watches each cell's voltage and the pack current.
Its job here	Protects the cells from over-charge, over-discharge, over-current and short circuit. It also acts as the **automatic cut-off**: when a cell reaches about 2.5–2.8 V it disconnects the load, which ends the discharge test. The current must flow through its **P−** output for this protection to work.
Why this one	Series Li-ion packs must have protection – one weak cell can be driven below a safe voltage. The 20 A rating gives a large safety margin over the ≈1.5 A test current. It must be a Li-ion (12.6 V) BMS, not a LiFePO4 one, whose voltage limits are different.

5. TP4056 charger modules (with protection)

1 A charge, 4.2 V, USB-C input, pads B+, B−, OUT+, OUT−

What it is	A small single-cell Li-ion charger that charges in two phases: constant current (1 A), then constant voltage (4.2 V) until the current falls. Red LED = charging, blue/green = full.
Its job here	Charges each cell **individually, out of the pack**, to exactly 4.2 V before every test.
Why this one	SOH is capacity, and capacity can only be measured by a complete discharge that starts from full – like filling a tank before measuring its size. Charging each cell separately also makes the three cells equal, so no single cell ends the test early. The protected version adds a safety cut-off for the cell.

B. The load (what drains the pack)

6. Cement power resistors R1 + R2

2 × 3.3 Ω, 25 W, in series = 6.6 Ω

What it is	Wire-wound resistors encased in ceramic cement, built to turn electrical energy into heat safely.
Its job here	They are the 'load' that discharges the pack: about 1.9 A when full, falling to about 1.2 A near empty (average ≈1.5 A). The energy leaves the pack as heat.
Why this one	In series the two give 6.6 Ω, which keeps the current near the 2 A used in the training data while each resistor dissipates only about 12 W of its 25 W rating. One alone (3.3 Ω) would overheat; the two in parallel (1.65 Ω) would draw a dangerous 7.6 A. Resistors are a low-cost substitute for an electronic constant-current load; the trade-off is that current falls slightly as the pack voltage drops.

7. SPDT toggle switch (3 pins)

rated ≥ 3 A; middle pin = common (C)

What it is	A single-pole switch: the middle pin connects to one of the two outer pins depending on the lever position.
Its job here	Starts and stops the discharge. The middle pin goes to PACK+, the pin that closes when ON goes to R1; the third pin is left unused.
Why this one	A switch is safer than connecting live wires by hand. Because the cells are always charged off-board, any current through the shunt must be a discharge, so the firmware treats all current as discharge (*DISCHARGE_ONLY_RIG* = *true*) and no second pole or GPIO27 jumper is needed.

C. Measurement

8. Current shunt

200 A / 75 mV (0.375 mΩ); big bolts carry current, small screws are for sensing

What it is	A very low, very precise resistance. Current through it produces a small voltage (Ohm's law: V = I × R) – about 0.6 mV at 1.5 A.
Its job here	Placed in the pack-negative line (PACK− to big bolt 1, load return to big bolt 2) so all the discharge current flows through it; the PZEM-017 measures the millivolts across its small screws and converts them to amps.
Why this one	A shunt measures current without breaking or loading the circuit noticeably and is the sensor the PZEM-017 is designed to use. Mounting it on the negative side keeps the sensing near 0 V, which the PZEM expects.

9. PZEM-017 DC power meter

0.05–300 V, Modbus RTU over RS485, 9600 baud 8N2; needs ≥ 7 V or USB 5 V

What it is	A ready-made DC energy meter: it measures voltage directly, current via the shunt, and calculates power and accumulated energy. Its measuring side is electrically isolated from its RS485 communication side.
Its job here	Measures the pack voltage (V+ to PACK+, V− to PACK−) and the discharge current (shunt sense wires) and answers the ESP32's requests for those values every 5 s.
Why this one	It is a low-cost, calibrated meter that gives voltage, current, power and energy in one module, which keeps the hardware simple – a key point of the project's 'low-cost' objective. Its isolation protects the ESP32 from the battery circuit. Its 1 Wh energy resolution is too coarse for one test, so the backend calculates energy itself from voltage × current × time.

10. MAX485 module (RS485 transceiver)

5 V; pins RO, RE, DE, DI, A, B, VCC, GND

What it is	A chip that converts the ESP32's serial signal (single-ended, 0–3.3 V) into RS485 (a differential pair, A and B) and back.
Its job here	Lets the ESP32 talk to the PZEM-017. DE and RE (joined, on GPIO4) switch it between sending a request and listening for the reply; DI carries data out (GPIO17), RO carries the reply in (GPIO16).
Why this one	The PZEM-017 only speaks RS485/Modbus; the ESP32 has no RS485 port. RS485 is also robust against electrical noise, which matters near switching loads and long wires.

11. Voltage divider (1 kΩ + 2 kΩ)

on the MAX485 RO → ESP32 GPIO16 line

What it is	Two resistors that scale a voltage down: Vout = 5 V × 2 kΩ ÷ (1 kΩ + 2 kΩ) ≈ 3.3 V.
Its job here	Reduces the MAX485's 5 V output to the 3.3 V level the ESP32 input can safely accept.
Why this one	ESP32 pins are 3.3 V devices; feeding them 5 V for long periods can damage them. Two resistors are the simplest, cheapest level shifter for a one-way signal.

D. Processing and communication

12. ESP32 Dev Module

dual-core 240 MHz, WiFi 2.4 GHz, 3 UARTs, 4 MB flash

What it is	A low-cost microcontroller board with built-in WiFi, programmed in Arduino C++.
Its job here	Every 5 s it requests a reading from the PZEM-017, divides voltage/power/energy by 3 (per cell), marks the current as discharge, adds a timestamp and sends it as JSON over WiFi to the laptop (HTTP POST). If WiFi drops it stores readings in its flash memory (SPIFFS) and sends them later.
Why this one	Built-in WiFi removes the need for a separate network module; a spare hardware serial port (UART2) talks to the MAX485; it is cheap and widely available, and well supported by the Arduino IDE.

13. Laptop + phone hotspot

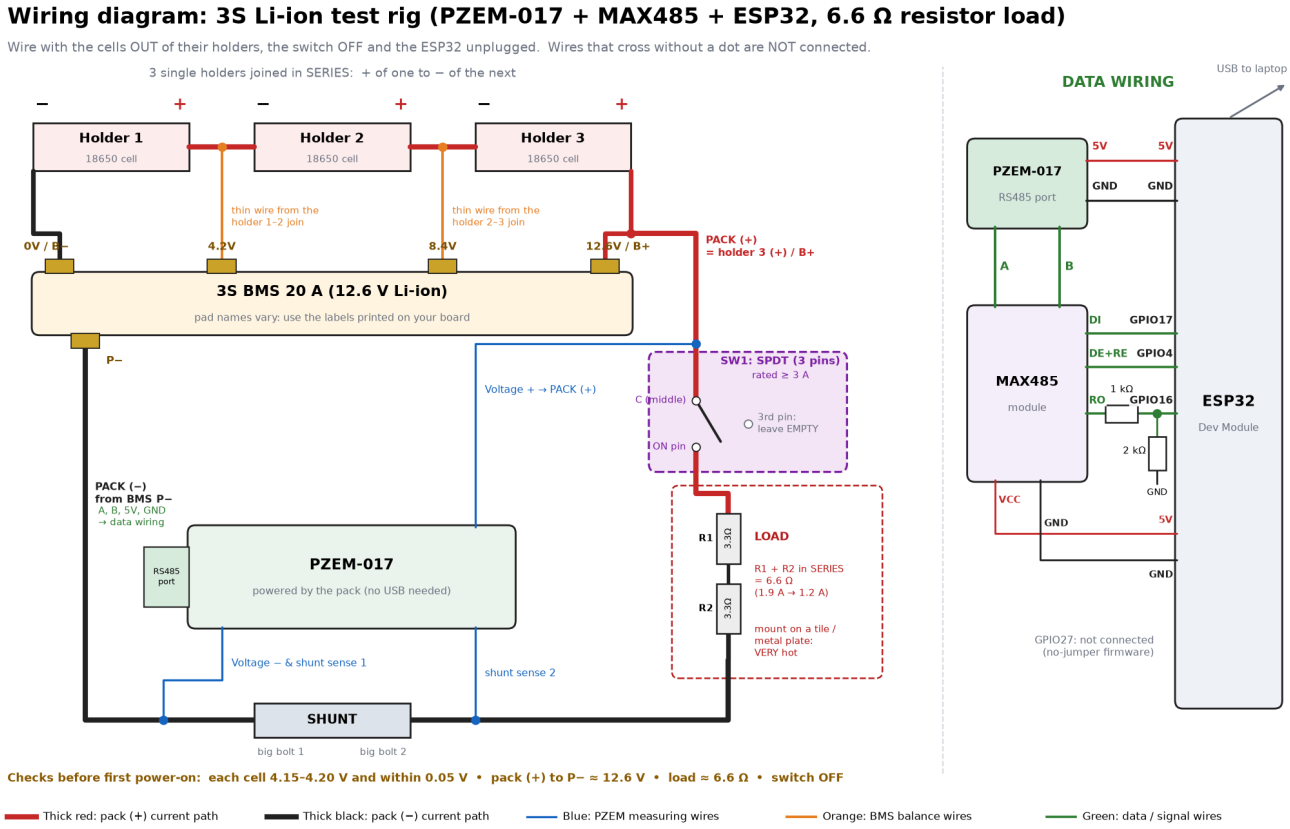
2.4 GHz WiFi, same network as the ESP32

What it is	The computer that runs the backend, database, ML models and dashboard; the hotspot provides the local network.
Its job here	Receives the ESP32's readings on port 8000 and shows the dashboard in a browser.
Why this one	Everything runs locally, with no cloud service or internet needed. A phone hotspot makes the setup portable for the defence. (The laptop's IP can change between sessions: SERVER_URL in the firmware must match it.)

E. Software components

Component	What it does	Why
ESP32 firmware (esp32_firmware_v2.ino)	Reads the PZEM over Modbus, applies per-cell scaling, sends JSON with a device timestamp, caches during WiFi loss.	Key settings: `SERVER_URL` (laptop IP), `CELLS_IN_SERIES = 3`, `DISCHARGE_ONLY_RIG = true`, `SHUNT_RANGE_CODE` (shunt size).
FastAPI backend (backend/main.py)	Stores every reading; a discharge starts when current falls below −0.5 A and ends when it rises above. It then computes mean voltage, mean current, mean power and energy (V × I × time), the charge delivered (mAh) and checks the discharge was complete before predicting.	The −0.5 A threshold ignores sensor noise. A partial discharge is recorded but not predicted, because the models only understand full discharges.
SQLite database (battery.db)	Keeps readings, cycles and predictions.	Zero set-up, single file; history survives restarts.
ML models (Random Forest + XGBoost)	Predict SOH and RUL from the four features; the dashboard shows each model and their average (ensemble).	Trained on NASA PCoE Li-ion data, tested on a battery they never saw (B0018): ensemble SOH R^2 0.975.
Streamlit dashboard	Live readings and charts, SOH gauge, RUL, alerts, and measured vs predicted SOH.	Turns the numbers into a clear maintenance decision; refreshes every 5 s.
Simulator (simulate_discharge.py)	Replays a real NASA discharge into the backend exactly as the ESP32 would.	Tests the software without hardware and gives a 20-second backup demo.

F. How it is all connected



Wiring diagram of the test rig (also in docs/Full_circuit_diagram.png; formal schematic in docs/Schematic_diagram.png).

G. Questions you are likely to be asked

Question	Short answer
Why three cells, not one?	The PZEM-017 needs ≥ 7 V. Three cells give 8.1–12.6 V; dividing by 3 keeps the inputs on the single-cell scale the models learned.
Why charge the cells before each test?	SOH is capacity, and capacity is only measured by a discharge from full to empty.
Why don't you discharge to 0 V?	A Li-ion cell is 'empty' at about 2.7 V; going lower damages it. The BMS cuts off automatically.
How do you know the prediction is right?	The backend measures the charge delivered (mAh). Measured SOH = mAh ÷ 2000 × 100, compared with the predicted SOH.
Why resistors instead of an electronic load?	Lower cost. The trade-off is a current that falls from ≈1.9 A to ≈1.2 A instead of a constant 2 A – a stated limitation.
Why no prediction after a short discharge?	Like a laptop's battery calibration, capacity needs a full discharge; a partial one would be misread as a worn battery.
Why not VRLA?	The models were trained on Li-ion; VRLA needs chemistry-specific normalisation and VRLA training data – future work.

Safety summary
Charge cells one at a time and never unattended • never discharge below 2.7 V per cell • always use the BMS and take PACK− from P− • the resistors exceed 100 °C – mount them on a tile or metal plate • stop at once if a cell gets hot, swells or smells.